

▣ 26 Correlation

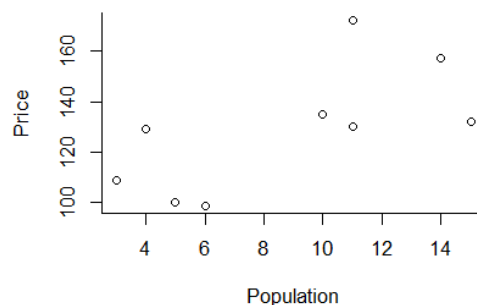
Example 1

An analysis is carried out to determine how, if at all, the price of houses is related to population density in the towns in which they are located. A selling price for a number of houses that have recently sold are noted as are the population of the towns in which they are located and are tabulated below.

Population (000)	3	14	15	6	10	11	4	11	5
Price (000)	109	157	132	99	135	130	129	172	100

Note that the two variables here are both of type scale so we begin by constructing a scatter diagram of the data

```
> par(bty="l") # Put an L rather than a square around the plot
> plot(Population, Price)
```



We can see some evidence that price tends to increase as population increases in a roughly linear fashion although the relationship is statistical rather mathematical.

We can quantify the magnitude of the linear relationship between the two variables using either the correlation r or the coefficient of determination r^2 . The former, r , must lie in the interval $[-1, 1]$ and so r^2 must lie in $[0, 1]$.

```
> cor(Population, Price) # calculates Pearson's correlation
[1] 0.6680599
> cor(Population, Price)^2
[1] 0.446304
```

Notes

1. The independent variable on the horizontal axis and the dependent variables on the vertical. In this case we can argue that price depends (if it does) on population and not the other way around. But in many other cases it is not so clear-cut (e.g. an individual's height and weight).
2. The term correlation is often used colloquially to mean related or associated, in any fashion. The correlation we are considering here is a measure of a linear (as in straight line) relationship

Interpreting r and r^2

Of interest is how close r is to 0 (no correlation) or 1 (fully correlated).

As $r = 0.688$ we have moderate correlation.

This notion of strong, or weak, correlation is of course subjective; it is better to look at r^2 , which has the following, more objective, interpretation.

In our sample price values, we can see (either from the table or the scatter diagram) that price varies, i.e. houses are not all sold for the same price.

Some of this variation can be explained by, or accounted for by, the fact that there is variation in the population of the towns.

In fact, we can partition the observed variation in house prices in to that which is explained by variation in town population, and all other factors combined.

The proportion of variation in price that can be explained by variation in population is given by $r^2 = 0.445$ or about 44.5% and the remaining variation is accounted for by factors other than population.

Example 2

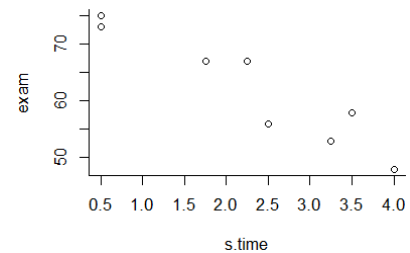
A study is carried out on a sample of ten children to investigate the relationship, if any, between screen time (the average time they spent per day using a smart phone / tablet) and their score in a diagnostic test.

s.time (hours)	0.5	1.75	2.5	2.25	3.5	3.25	4	0.5	0.5
exam (x 100)	75	67	56	67	58	53	48	73	75

```
> plot(s.time, exam)
```

The scatter plot suggests an inverse relationship.

Also, the relationship appears to be stronger than in the previous example.



```
> cor(s.time, exam)
[1] -0.9521641
> cor(s.time, exam)^2
[1] 0.9066164
```

The correlation $r = -0.95$ This is negative because the variables are negatively related.

Thus, we assess the strength of the relationship in this case by examining how close r is to -1 . The value of -0.95 is consistent with our observation that the relationship is quite strong.

$r^2 = 0.91$ we can explain 91% of the variation observed in exam scores by looking at variation in screen time.

Alternatively, we might say that the variation in screen time would be reduced by 91% if all children has identical screen times.

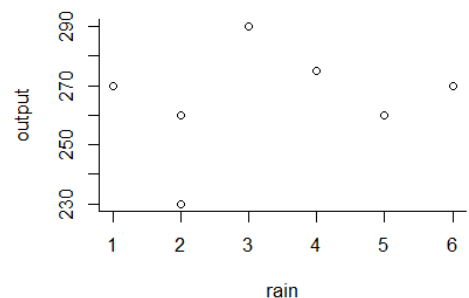
Example 3

A study is carried out in a production facility to determine if inclement weather affects productivity. A random sample of shifts is taken from the last 6 months and the number of units manufactured in that shift observed. A meteorological database is consulted to determine the amount of rain that fell during that shift.

Rain	2	3	1	4	5	6	2
Output	260	290	270	275	260	270	230

```
> plot(s.time, exam)
cor(rain, output)
[1] 0.2254837
> cor(rain, output)^2
[1] 0.0508429
```

The scatter plot suggests no relationship.
Rainfall and productivity are not related.
Rainfall and productivity are independent



We might reasonably have guess that the two variables here might be independent.

But still, do did not obtain a correlation value of 0, $r = 0.225$

And $r^2 = 0.05$, which suggests that 5% of variation in productivity is explained by variation in rain fall.

N.B. You will almost always obtain a non-zero correlation because of sampling error.

Clearly, we need a means to determine when there is a real relationship between two variables, even if the relationship is weak.

We do so by conducting a statistical test to see if the correlation differs from 0.

Consider the house price v town population example again

H_0 : The correlation between price and population is 0.

H_A : The correlation between price and population is not 0.

```
> cor.test(Population, Price)

Pearson's product-moment correlation

data: Population and Price
t = 2.3754, df = 7, p-value = 0.04922
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 0.007078894 0.922772350
sample estimates:
      cor
0.6680599
```

As with all our tests, we reject the null hypothesis if $p < 0.05$, as is the case here.

Thus, we conclude that there is a relationship between the two variables?

The test also gives a confidence interval estimate for the true correlation – note that the interval is very wide.

Test assumptions

It is assumed that the two variables have what is called a joint normal distribution.

This assumption can be tested using the Henze-Zirkler test – this is in the package `mvnTest`

Note that we do pass one argument, and not two, to this function. This one argument should be a data frame, containing the variables to be tested. The function `data.frame` will create a data frame consisting of all the variables passed as arguments.

```
HZ.test(data.frame(Population, Price))
      Henze-Zirkler test for Multivariate Normality

data : data.frame(Population, Price)

HZ           : 0.1447857
p-value      : 0.9564112

Result  : Data are multivariate normal (sig.level = 0.05)
```

So, we can be confident that the correlation is statistically significantly different from zero.

Consider the screen time v exam score example again

```
> cor.test(s.time, exam)

Pearson's product-moment correlation

data:  s.time and exam
t = -8.2438, df = 7, p-value = 7.521e-05
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 -0.9901571 -0.7834750
sample estimates:
      cor 
-0.9521641

> HZ.test(data.frame(s.time, exam))
      Henze-Zirkler test for Multivariate Normality

data : data.frame(s.time, exam)

HZ          : 0.3370962
p-value     : 0.3973684

Result  : Data are multivariate normal (sig.level = 0.05)
```

Again, we have a statistically significant correlation and again the assumptions are satisfied, In this example the confidence interval estimate is considerably narrower.

Consider the rain v production output example again

```
> cor.test(rain, output)

Pearson's product-moment correlation

data:  rain and output
t = 0.51752, df = 5, p-value = 0.6269
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 -0.6354805  0.8365018
sample estimates:
      cor 
0.2254837
```

In this case, we see that the correlation is not statistically significantly different from zero.

Consequently, we do not need to check for joint or multivariate normality.

Example

In the customer survey data, is there a (linear) correlation between the customers age and the amount they spend?

The construction of a scatter plot is left as an exercise

Note that unlike our other examples, there are some missing values in our data here. If we want to calculate the correlation (without running the test for significant) we can do so with

```
> cor(q2_age, q4_spend, use="pairwise.complete")
[1] 0.6167301
```

However, we do not need the additional argument if we run the test.

```
cor.test(q2_age, q4_spend)

Pearson's product-moment correlation

data:  q2_age and q4_spend
t = 5.7573, df = 54, p-value = 4.165e-07
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 0.4223009 0.7569068
sample estimates:
      cor
0.6167301
```

So, there is a statistically significant linear relationship between age and spend. And as $r^2 = 0.38$, we can say that 38% of the variation in spend can be explained by variation in age.

We do encounter a complication because of the missing values when we try to check the assumptions. The test for normality requires that there be no missing values.

We can strip away the missing values using the function `is.na( )` as follows

```
> age = q2_age[!is.na(q2_age) & !is.na(q4_spend)]
> spend = q4_spend[!is.na(q2_age) & !is.na(q4_spend)]
> HZ.test(data.frame(age, spend))
      Henze-Zirkler test for Multivariate Normality

data : data.frame(age, spend)

HZ          : 0.5900324
p-value     : 0.301089

Result  : Data are multivariate normal (sig.level = 0.05)
```

So, the assumption of normality is fine.

□ Silly questions!

It is not uncommon to find novice users of statistics report that they have found a statistically significant correlation in some research that they are conducting when such a finding was entirely unremarkable!

For example, it might be reported that there is a statistically significant, positive correlation between the weights and heights of players on a football team. Or a statistically significant, positive correlation between the size of digital movies and the time it takes to download them.

Or it might be reported that there is a statistically significant, negative correlation between the distance travelled by a car on a number of trips and the amount of fuel remaining in the fuel tank. Or, for a family renovating their home, a statistically significant, negative correlation between time and remaining funds.

Remember that the null hypothesis for tests of correlation is that the correlation is zero so a statistically significant finding only establishes that there is some (non-zero) correlation between the two variables.

In all of the examples above we would fully expect the variables to be correlated so there is nothing remarkable about establishing such a finding. The time it takes to download a file will clearly tend to be higher for larger files – it is somewhat silly to ask if your sample data suggests such a relationship. In fact, it would be far more remarkable if you could show that these two variables are *not* related.